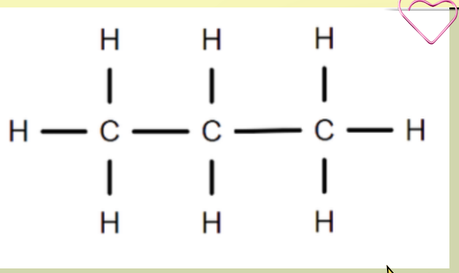
PROPANE

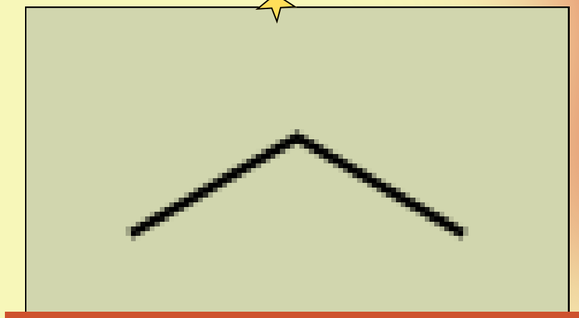
COMPOUND 1

level: foundation

formula:	C3H8
condensed:	CH3CH2CH3
IUPAC name:	PROPANE
common:	DIMETHYLMETHANE
chE link	fuel processing



lewis structure



bond-line structure (ketcher)



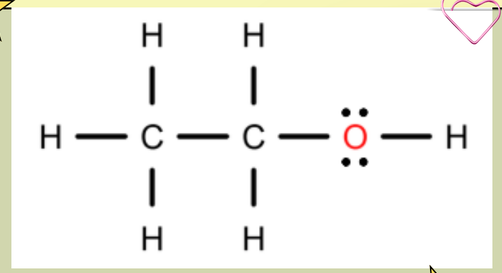
atom count verified
valence checked
same connectivity

ETHANOL

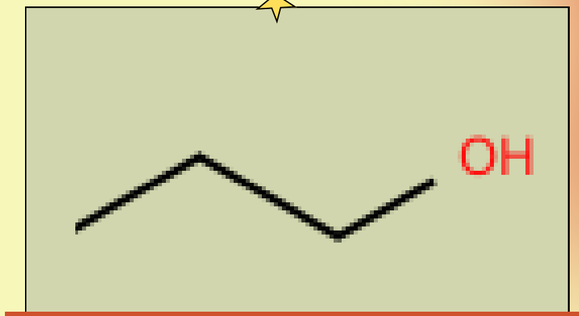
COMPOUND 2

level: foundation

formula:	C2H6O
condensed:	CH3CH2OH
IUPAC name:	ETHANOL
common:	ETHYL ALCOHOL
chE link	fermentation



lewis structure



bond-line structure (ketcher)



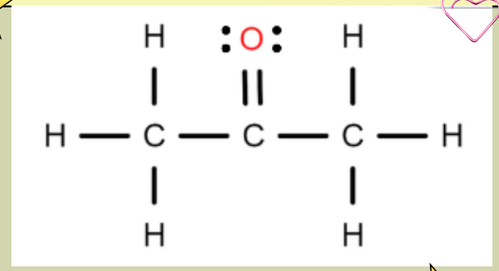
atom count verified
valence checked
same connectivity

ACETONE (PROPAN-2-ONE)

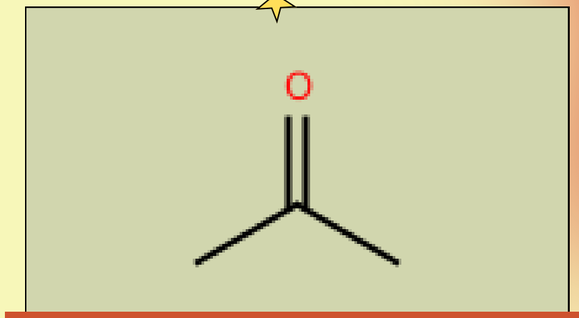
COMPOUND 3

level: foundation

formula:	C3H6O
condensed:	CH3COCH3
IUPAC name:	2-PROPANONE
common:	ACETONE
chE link	industrial solvent



lewis structure



bond-line structure (ketcher)

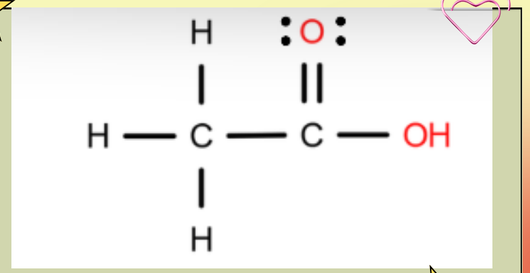


atom count verified
valence checked
same connectivity

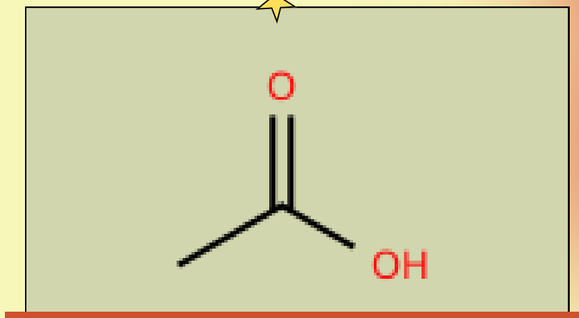
ACETIC ACID (ETHANOIC ACID)

COMPOUND 4

level: developing



lewis structure



bond-line structure (ketcher)

formula:	C2H4O2
condensed:	CH3COOH
IUPAC name:	ETHANOIC ACID
common:	ACETIC ACID
chE link	acid processing



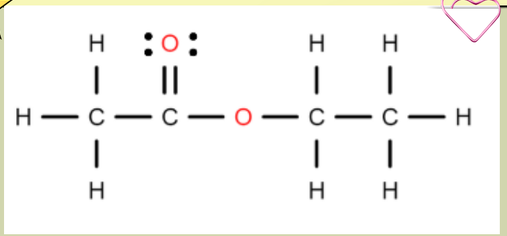
atom count verified
valence checked
same connectivity

ETHYL ACETATE (ETHYL ETHANOATE)

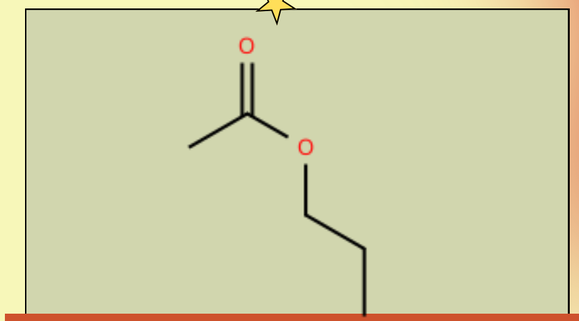
COMPOUND 5

level: developing

formula:	C4H8O2
condensed:	CH3COOCH2CH3
IUPAC name:	ETHYL ETHANOATE
common:	ETHYL ACETATE
chE link	esterefication



lewis structure



bond-line structure (ketcher)

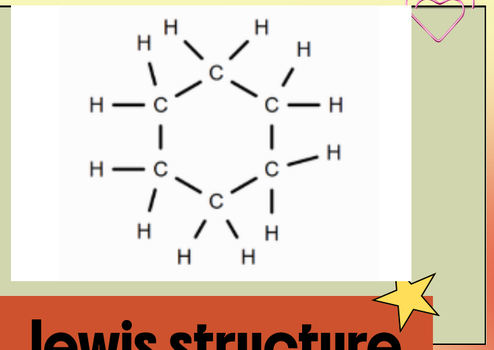


atom count verified
valence checked
same connectivity

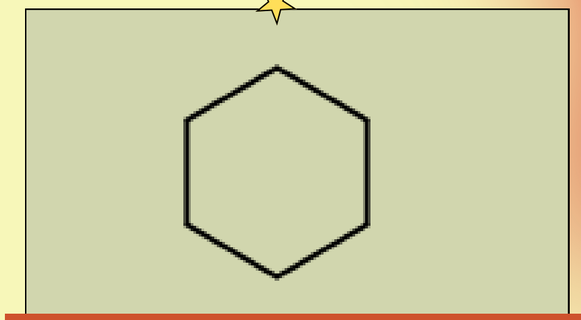
CYCLOHEXANE
COMPOUND 6

level: developing

formula:	C6H12
condensed:	C6H12
IUPAC name:	CYCLOHEXANE
common:	
chE link	petrochemical intermediate



lewis structure



bond-line structure (ketcher)

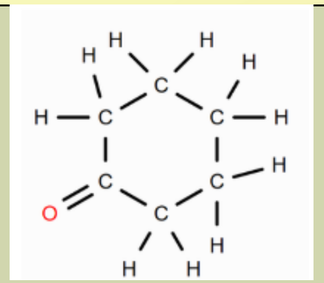


atom count verified
valence checked
same connectivity

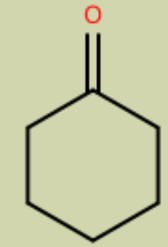
CYCLOHEXANONE
COMPOUND 7

level: developing

formula:	C6H10O
condensed:	C6H10O
IUPAC name:	CYCLOHEXANONE
common:	
chE link	nylon manufacturing



lewis structure



bond-line structure (ketcher)



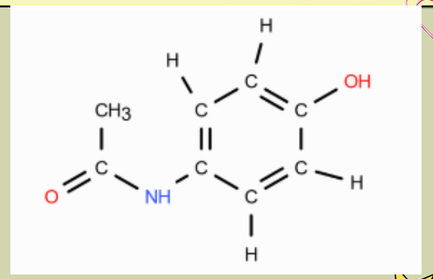
atom count verified
valence checked
same connectivity

ACETAMINOPHEN (PARACETAMOL)

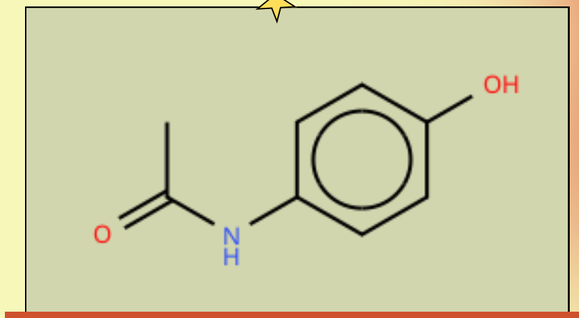
COMPOUND 8

level: challenge

formula:	C8H9NO2
condensed:	CH3CONHC6H4OH
IUPAC name:	4-ACETAMIDOPHENOL
common:	PARACETAMOL
chE link	pharmeceutical synthesis



lewis structure



bond-line structure (ketcher)



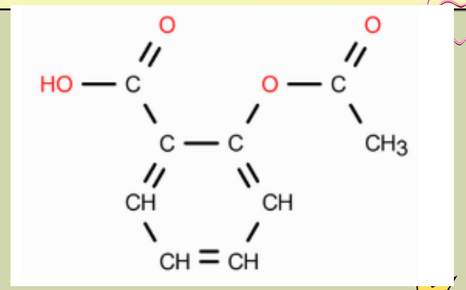
atom count verified
valence checked
same connectivity

ASPIRIN (ACETYLSALICYLIC ACID)

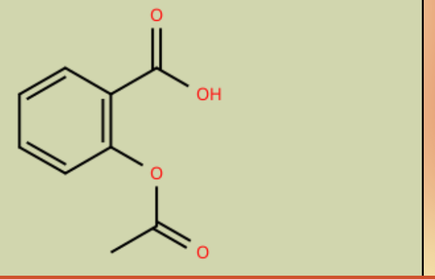
COMPOUND 9

level: challenge

formula:	C9H8O4
condensed:	CH3COOC6H4COOH
IUPAC name:	2-ACETOXYBENZOIC ACID
common:	ASPIRIN
chE link	pharmeceutical reaction



lewis structure



bond-line structure (ketcher)



atom count verified

valence checked

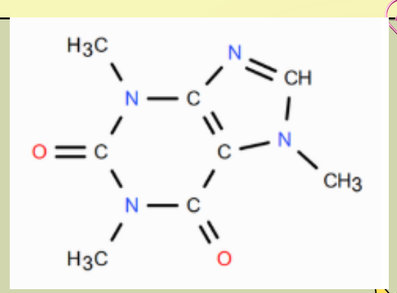
same connectivity

CAFFEINE

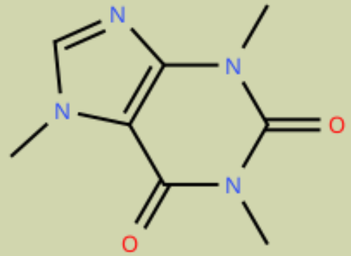
COMPOUND 10

level:
challenge

formula:	C8H10N4O2
condensed:	(CH3)2(C4N2O2)(CN2H)(CH3)
IUPAC name:	2-ACETOXYBENZOIC ACID
common:	ASPIRIN
chE link	extraction and separations



lewis structure



bond-line structure (ketcher)



atom count verified
valence checked
same connectivity